

Muhammad Arham Akmal

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WORK EXPERIENCE

TeReSol

Jun 2024 – Aug 2024

Software Engineer Internee

Islamabad, Pakistan

- Developed and maintained a Learning Management System (LMS) using Vue.js, utilizing Vuex for efficient state management and integrating XState for managing complex state transitions and workflows.
- Participated in agile development processes, including sprint planning, stand-ups, and retrospectives.
- Collaborated with a team of developers to implement new features and enhance existing functionalities.

FAST National University of Computing and Emerging Sciences

Jan 2024 – May 2024

Lab Demonstrator – Computer Organization and Assembly Language

Islamabad, Pakistan

- Facilitated lab sessions for undergraduate students, clarifying concepts in computer organization and assembly language programming.
- Provided individualized support on COAL assignments, debugging, and project development.

EDUCATION

FAST National University of Computing and Emerging Sciences

2021-2025

BS Computer Science - 7th Semester

Islamabad,

Pakistan

- Studied Programming fundamentals, Object Oriented Programing, Data Structures, Design and Analysis of Algorithms, Computer Organization and Assembly Language, Intro to AI, Parallel and Distributive Computing, Computer Networks, Deep learning, NLP
- I was an active member of social societies and managed large-scale events like NaSCon, Job Fairs, and Alumni Homecoming.

The Science School

2015-2021

O/A levels

Islamabad, Pakistan

- Studied Design and Technology, physics, Mathematics, Chemistry, and Computer Science.

Personnel Projects

- **Glaucoma detection system** (TensorFlow, Image Processing, and Python): Created a glaucoma detection system which achieved 90% accuracy in detecting glaucoma.
 - **Predicting and Generating Video Sequences Using Deep Learning** (Python, Pytorch, OpenCV, Streamlit, Gradio, Flask, MoviePy): Developed a video prediction model to generate future video frames based on short input sequences, using the UCF101 dataset of human activity videos.
 - **Machine Translation: Urdu-to-English** (Python, PyTorch, Moses BLEU, Matplotlib, NLTK.): Designed and implemented an attention-based encoder-decoder model to translate Urdu sentences into English using a custom parallel dataset.
 - **Flight Delay Time Prediction Using Ensemble Learning** (Python, Pandas, NumPy): Developed an ensemble learning model to predict flight delay times using three machine learning algorithms implemented from scratch: SVM, Decision Trees and Logistic Regression.
 - **NLP N-Gram Movie Review Classifier** (Python, Unigram, Bigram, Trigram): Built a N-Gram model from scratch to predict the next word without the use of external libraries. Applied Bayesian classification to label movie reviews generated by the language model.
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